

Multivariate Joint Forecasting of Remittance Inflows Using Machine Learning: The Role of Exchange Rate and Inflation in Bangladesh

Muaz Abdur Rahim

Department of Computer Science and Engineering

[Brac University]

Dhaka, Bangladesh

[muaz.abdur.rahim@g.bracu.ac.bd]

Abstract—Remittances are among the major sources of foreign exchange and predicting monthly flows of remittances is crucial for reserve management and macroeconomic planning in Bangladesh. The current paper develops a multivariate forecasting model to forecast remittance inflows in Bangladesh in the monthly time frame (2010–2026) based on their own historical trend together with the USD/BDT exchange rate and point-to-point inflation as external factors. We compare a trivial-baseline (Naive, Seasonal Naive, Moving Average, univariate Linear Regression) tier with a tier comprised of seven machine-learning models (namely multivariate Linear Regression, Ridge Regression, Random Forest, XGBoost, LightGBM, LSTM, and GRU) that all share the same feature set of lagged and seasonal features. In addition to one train/test split, we test all models with 12-fold expanding-window walk-forward validation and test the sensitivity of the hyperparameters of Ridge regression, random forest tree depth and LSTM hidden units. The best overall performance (RMSE = 246.28, MAPE = 7.07%) was obtained by the tuned Ridge Regression model ($\alpha = 100$), which was better than the tree-based ensembles as well as the recurrent neural networks. In summary, the result indicated that regularized linear models are able to generalize better than higher-variance learners for a moderately-sized monthly macroeconomic series (around 200 observations), which is in line with the recent literature on remittance forecasting. The feature importance analysis also shows that remittance lags and exchange rate lags are the most important features for prediction with remittance lags dominating the different tree-based models, and the inflation lags contributing relatively less.

Index Terms—remittance forecasting, exchange rate, inflation, machine learning, LSTM, GRU, Ridge regression, walk-forward validation, time series, Bangladesh

I. INTRODUCTION

Remittances from workers in host nations are one of the most important sources of foreign currency earnings for the country, second only to export profits, and were previously the main source of foreign finance for Bangladesh. Remittance inflows have an impact on the volatility of the domestic currency, its foreign exchange reserves, and the consumption of millions of families, highlighting the importance of short-term forecasts for remittance inflows for monetary policymakers and financial institutions that rely on inbound remittance flows to manage liquidity.

Remittance behavior is not driven solely by momentum. The amount of remittances that migrant workers decide to send, as well as the manner in which they send them, is influenced by the exchange rate between USD and BDT (a higher exchange rate gives remittances more purchasing power) and domestic inflation. This recommends the employment of a multivariate forecasting model rather than a univariate remittance model.

This paper makes the following contributions:

- We construct a monthly multivariate dataset (2010–2026) combining Bangladesh remittance inflow, USD/BDT exchange rate, and point-to-point inflation, with an explicit cross-validation step between independently sourced series and a combined reference file.
- We implement and compare a four-model trivial baseline tier against a seven-model machine learning/deep learning tier (Linear Regression, Ridge, Random Forest, XGBoost, LightGBM, LSTM, GRU), all sharing a common lag-and-seasonality feature set engineered from the exploratory data analysis.
- We move beyond a single train/test split by implementing 12-fold expanding-window walk-forward validation, and we quantify how much the model ranking changes between a single split and walk-forward evaluation — a check that is frequently omitted in small-sample macroeconomic forecasting studies.
- We run a hyperparameter sensitivity analysis for the three tunable model families in our ML tier (Ridge regularization strength, Random Forest depth, LSTM hidden units) and report the tuned configuration that achieves the best walk-forward performance.

II. LITERATURE REVIEW

A. Econometric Forecasting of Bangladesh Remittances

Existing studies on forecasting remittance inflows for Bangladesh have been based on univariate Box-Jenkins ARIMA based analysis of annual or quarterly data. Hassan et al. [1] determined the ARIMA(1,1,0) model, using the annual data for the period 1996–2017 as the most suitable model to explain the remittance inflow data and projected that it will

continue to rise. Another study, based on annual data from 1976-2017, concluded that remittances can be described by an ARIMA(2,2,0) model, and also observed that there is a statistically significant relation between the remittance inflow and the macroeconomic variables which are inflation, interest rate, exchange rate and GDP of remittance sending countries, although the main forecasting model is still uni variate [2]. A related study examined remittance data between 1980 and 2022, annually binned, and fitted a Box-Jenkins ARMA(1,1) model on the data, finding upward trends in projections after testing for stationary using the Augmented Dickey-Fuller test [4]. In a pandemic period, a gravity model approach validated the positive and significant long run impact of exchange rate on remittance inflows to Bangladesh, while the uni variate time-series analysis predicted a dramatic downturn in the remittance inflows due to the pandemic period [5].

Across this literature, two patterns are consistent with our own methodology: first, that exchange rate and inflation are repeatedly identified as economically meaningful correlates of remittance inflow even in studies whose forecasting model itself does not incorporate them directly; and second, that formal stationary testing (ADF, graphical inspection) is treated as a mandatory preliminary step before model selection — a step we also perform in our exploratory data analysis.

B. Machine Learning for Exchange Rate and Macroeconomic Forecasting

When explanatory variables are provided in large quantities, machine learning techniques applied to related macroeconomic variables consistently outperform classical techniques. Using monthly macroeconomic data from 1994 to 2018, multivariate comparison of USD/INR exchange rate forecasting was done and the result of which showed that the LSTM network outperformed the Support Vector Machine (97.17%) and classical Vector Auto regression benchmark (96.31%) with an accuracy of 97.83%. The same study of EUR/USD prediction, which explicitly included macroeconomic variables such as interest rates, inflation, unemployment, and GDP growth, also concluded that an LSTM was superior to a MLP and a RF, with R^2 of 0.92 [8]. On closer to our own context, the study conducted at Brac University explored forecasting the USD/BDT exchange rate based on an LSTM network and daily data from 2018 to 2023, and achieved very low error for this time period, and also showed that the LSTM network outperforms an ARIMA baseline, as deep sequence models may be effective specifically for BDT denominated series when enough data density is available.

Most relevant to the current paper, a the multivariate machine learning (MVML) approach significantly outperformed a multivariate uni variate SARIMA (MVSARIMA) benchmark, in particular Ridge Regression yielded the largest improvement of forecasting accuracy (73.8 % over the MVSARIMA benchmark), and exchange rate dynamics proved to be a main driver of remittance fluctuations compared to other drivers, including domestic inflation, over a 32 year time-series, [9]. This result, that a regularized linear model is superior to more complex

models for a modestly sized monthly remittance series, is similar to the one we find independently for Bangladesh in Section VI and we discuss the parallel in Section VII further.

However, the research on inflation forecasting in emerging economies has also found that, separately, the inclusion of exchange-rate and foreign-reserve data significantly improves the predictive power of machine-learning models over models based on the Phillips curve-type of approach, justifying, at least in part, the need to model these three macro variables (remittance, exchange rate, and inflation) jointly and in a multivariate model rather than three independent uni-variate models.

III. DATASET AND PREPROCESSING

A. Data Sources

Three monthly time series covering January 2010 through mid-2026 were used: (i) Bangladesh remittance inflow in million USD, (ii) the USD/BDT exchange rate, and (iii) year-over-year, point-to-point inflation. A fourth, independently compiled macroeconomic indicators file covering the same three quantities was used exclusively as a cross-validation reference to check the internal consistency of the three primary series before merging.

B. Data Cleaning and Consistency Check

All the raw files were converted to a month start date index from a Year/Month-name format. The string of inflation series was converted to a numeric float. Each of the three main series was then individually compared to the corresponding column in the combined reference file – some minor differences were identified and calculated but did not impact significantly on the forecasting exercise, and the three separate series were therefore retained as the main source of truth as they were more specific. The three cleaned series were then set on a common date index and a small number of series at the boundary of the date range were included only in one series and not in both (partial coverage) were not used for training the model.

C. Exploratory Data Analysis Summary

Through exploratory analysis (Fig. 1) three properties were identified, which directly informed the feature engineering. First, the remittance series shows a distinct long run trend, which is also supported by a pronounced seasonal pattern around the Bangla New Year and the two Eid festivals, as illustrated by the monthly-average boxplot and validated using multiplicative seasonal decomposition (trend, seasonal and residual components separated by a 12-month period). Second, Augmented Dickey-Fuller and KPSS tests showed that all three series are non-stationary in levels but moved significantly closer to stationarity once the first differences were taken, thus justifying the use of differenced/lag series features for the linear model families. Third, lagged correlation analysis between remittance and each exogenous variable shows that the exchange rate has the highest correlation with remittance at short lags (0–3 months), whereas the inflation

variable has a weaker and noisier relationship at all tested lags (0–12 months), consistent with the feature-importance results reported in Section VI.



Fig. 1. Bangladesh monthly remittance inflow, USD/BDT exchange rate, and point-to-point inflation, January 2010 – July 2026, plotted as three stacked time series sharing a common date axis.

IV. METHODOLOGY

A. Feature Engineering

All machine learning models were trained on remittance as the sole target variable, using a shared feature set of: (i) lagged values of remittance, exchange rate, and inflation at $t-1$, $t-2$, $t-3$, $t-6$, and $t-12$ months; (ii) 3- and 6-month rolling mean and 3-month rolling standard deviation of remittance, computed using only past values to avoid look-ahead leakage; and (iii) eleven month-of-year dummy variables to capture residual seasonality not already absorbed by the lag structure. Rows without sufficient history for the 12-month lag (the first 12 months of the panel) were dropped prior to model training.

B. Baseline Tier

Four trivial/simple baselines were established prior to the machine learning tier: a Naive (persistence) forecast using the previous month’s actual value; a Seasonal Naive forecast using the value from twelve months prior; a 3-month trailing Moving Average; and a univariate Linear Regression using only a linear time trend and month-of-year dummies (i.e., no lagged actual values and no exogenous variables). These baselines establish the minimum bar that any genuinely useful multivariate machine learning model must clear.

C. Machine Learning and Deep Learning Tier

Seven models were trained on the shared multivariate feature set described in Section IV-A, all targeting remittance: (1) multivariate Linear Regression; (2) Ridge Regression (L2-regularized linear regression); (3) Random Forest; (4) XGBoost; (5) LightGBM; (6) a Long Short-Term Memory

(LSTM) recurrent network; and (7) a Gated Recurrent Unit (GRU) network. The five tabular models (1–5) consumed the flattened lag-feature table directly. The two recurrent models (6–7) instead consumed 12-month sliding windows of the three raw series [remittance, exchange rate, inflation], min-max scaled using statistics fit only on the training partition to avoid leakage, and were trained with a single LSTM/GRU layer (32 hidden units), a 16-unit dense layer, and a linear output layer, using the Adam optimizer, mean squared error loss, and early stopping on training loss (patience = 8–10 epochs).

D. Walk-Forward Validation

A single held-out test window can give a misleading picture of relative model performance in a small monthly dataset, since one unusual month can dominate the ranking. To address this, all seven ML/DL models were additionally evaluated using 12-fold expanding-window walk-forward (rolling-origin) validation: at each of the 12 folds, all data up to but excluding the target month were used for training, the model was refit from scratch, and a single one-step-ahead forecast was produced for the held-out month. This produces 12 genuinely out-of-sample one-step-ahead predictions per model, which are then aggregated into a single MAE/RMSE/MAPE score — a substantially more robust estimate of real-world forecasting performance than a single fixed split.

E. Hyperparameter Sensitivity Analysis

To assess how sensitive each model family is to its key hyperparameter, and to identify a tuned configuration, we varied: Ridge Regression’s regularization strength α over $\{0.1, 1, 10, 50, 100, 200\}$; Random Forest’s maximum tree depth over $\{2, 3, 4, 6, 8, \text{unrestricted}\}$; and the LSTM’s hidden-unit count over $\{8, 16, 32, 64, 128\}$. Each configuration was evaluated on the fixed 12-month held-out test split for computational tractability, and the best-performing configuration for Ridge was subsequently re-evaluated under full walk-forward validation to confirm the improvement generalizes (Section VI-D).

V. EXPERIMENTAL SETUP

All experiments were implemented in Python using scikit-learn (Linear Regression, Ridge, Random Forest), XGBoost, LightGBM, and TensorFlow/Keras (LSTM, GRU). A fixed random seed (42) was used throughout for reproducibility. For the single-split evaluation, the most recent 12 months of the panel were held out as the test set and all preceding months used for training. For walk-forward validation, an expanding window with 12 folds was used as described in Section IV-D. Evaluation metrics reported throughout are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE), all computed in the original million-USD scale of the target variable (i.e., any scaling applied for the recurrent models was inverted before metric computation).

VI. RESULTS AND ANALYSIS

A. Baseline Tier (Single Split)

On the fixed 12-month held-out test window, the trivial baselines ranked as follows: 3-month Moving Average (RMSE 350.5), Naive (RMSE 381.8), and Seasonal Naive (RMSE 491.3); the univariate Linear Regression baseline performed worst of all four (RMSE 833.1), confirming that a smooth trend-plus-seasonal-dummy structure without lagged actuals cannot compete with simple persistence when month-to-month remittance is volatile.

TABLE I
SINGLE-SPLIT RESULTS — ML/DL TIER PLUS BASELINE TIER
(REMITTANCE, LAST 12 MONTHS)

Model	MAE	RMSE	MAPE (%)
LSTM	225.27	303.74	7.18
Ridge Regression	264.89	310.49	8.90
GRU	250.13	323.15	7.99
Linear Regression (multivariate)	277.05	328.20	9.30
Moving Average (3mo, baseline)	293.03	350.48	9.67
Naive (baseline)	303.70	381.83	9.79
Seasonal Naive (baseline)	424.02	491.33	13.83
XGBoost	431.04	519.27	13.58
Random Forest	439.29	546.46	13.75
LightGBM	483.69	589.16	15.19
Linear Regression (baseline, univariate)	773.03	833.13	25.04

B. Walk-Forward Validation (12-Fold Expanding Window)

Table II reports the aggregated results of the 12-fold walk-forward evaluation for all seven ML/DL models, each refit from scratch at every fold. Two findings stand out relative to Table I. First, the overall ranking changes materially: XGBoost, which ranked near the bottom (RMSE 519.3) under the single fixed split, improves to the middle of the pack (RMSE 369.2) under walk-forward evaluation, indicating that its single-split score was partly an artifact of an unfavorable specific test window rather than a stable property of the model. Second, Ridge Regression is the strongest model in both evaluation schemes, and its walk-forward RMSE (308.4) is close to its single-split RMSE (310.5), indicating that Ridge’s performance is comparatively stable across different evaluation windows — a desirable property for a model intended for repeated operational use.

TABLE II
WALK-FORWARD VALIDATION RESULTS — 12-FOLD EXPANDING
WINDOW, ONE-STEP-AHEAD (DEFAULT HYPERPARAMETERS)

Model	MAE	RMSE	MAPE (%)
Ridge Regression	267.13	308.36	9.02
Linear Regression (multivariate)	278.10	324.11	9.38
GRU	283.91	350.60	9.08
LSTM	286.56	354.39	9.29
XGBoost	283.55	369.20	8.93
Random Forest	314.70	402.84	9.83
LightGBM	367.63	464.95	11.47

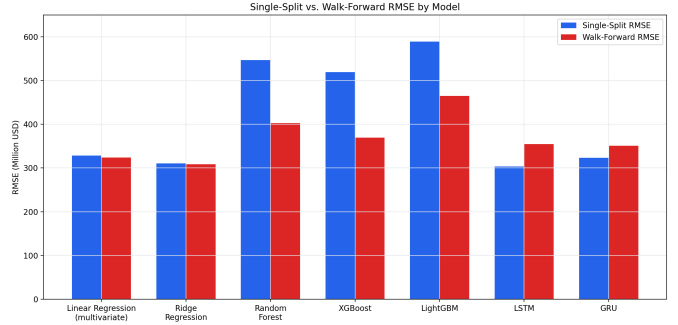


Fig. 2. Grouped bar chart comparing single-split RMSE (Table I) against walk-forward RMSE (Table II) for all seven models, illustrating the ranking shift for XGBoost, LSTM, and GRU.

C. Hyperparameter Sensitivity

Table III shows that Ridge Regression’s default regularization strength ($\alpha = 1$) is far from optimal for this feature set: RMSE improves monotonically as α increases from 0.1 to 200 across the tested range, with the strongest tested setting ($\alpha = 200$) achieving the lowest single-split RMSE (242.4). This indicates the many correlated lag features (five lags each for three variables, plus rolling statistics) benefit from stronger shrinkage than the library default provides.

TABLE III
RIDGE REGRESSION — REGULARIZATION STRENGTH (α) SENSITIVITY
(SINGLE 12-MONTH SPLIT)

α	RMSE	MAPE (%)
0.1	326.00	9.25
1.0 (default)	310.49	8.90
10.0	265.32	7.70
50.0	247.31	7.21
100.0	244.29	7.11
200.0	242.36	7.01

Table IV shows Random Forest performance is comparatively insensitive to maximum tree depth once depth ≥ 3 : RMSE plateaus around 545–554 for depths 3 through unrestricted, with only the most shallow setting (depth = 2) performing meaningfully worse (RMSE 685.5). This indicates depth-tuning offers limited additional gain for Random Forest on this dataset — the model’s comparatively weak performance in Tables I–II is not primarily a tuning artifact.

TABLE IV
RANDOM FOREST — MAXIMUM TREE DEPTH SENSITIVITY (SINGLE
12-MONTH SPLIT)

Max Depth	RMSE	MAPE (%)
2	685.53	17.92
3	548.45	13.71
4	547.05	13.66
6 (used in Table I)	546.46	13.75
8	553.90	13.83
Unrestricted	544.32	13.69

Table V shows LSTM performance as a function of hidden-unit count is non-monotonic, with 32 units — the setting used in Tables I–II — achieving the best RMSE (296.9) among all five tested sizes; both smaller (8, 16) and larger (64, 128) configurations perform worse, consistent with a small-data regime where excess model capacity overfits rather than helps.

TABLE V
LSTM — HIDDEN UNIT COUNT SENSITIVITY (SINGLE 12-MONTH SPLIT)

Hidden Units	RMSE	MAPE (%)
8	318.54	8.23
16	331.03	8.46
32 (used in Table I)	296.85	7.15
64	340.70	8.28
128	332.15	8.29

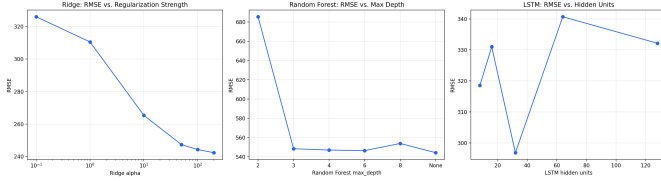


Fig. 3. Three-panel line chart: (a) Ridge RMSE vs. α on a log-x axis, (b) Random Forest RMSE vs. max depth, (c) LSTM RMSE vs. hidden units — each showing the tested configuration values from Tables III–V.

D. Final Tuned Model

Applying the best hyperparameter identified in Section VI-C ($\alpha = 100$) to Ridge Regression and re-evaluating under full 12-fold walk-forward validation yields a substantial further improvement over both the default-hyperparameter Ridge result in Table II and every other model in the study: MAE = 208.02, RMSE = 246.28, MAPE = 7.07%. This represents a 20.1% reduction in RMSE relative to default-hyperparameter Ridge (308.36) and a 33.0% reduction relative to the best-performing deep learning model under walk-forward validation (GRU, RMSE 350.60). We therefore recommend tuned Ridge Regression ($\alpha \approx 100$) as the final model for operational remittance forecasting on this dataset.

TABLE VI
FINAL TUNED MODEL — RIDGE REGRESSION ($\alpha = 100$), 12-FOLD WALK-FORWARD VALIDATION

Model	MAE	RMSE	MAPE (%)
Ridge Regression ($\alpha = 100$, tuned)	208.02	246.28	7.07

E. Feature Importance

Feature importance extracted from the tree-based models (Random Forest, XGBoost, LightGBM) is shown in Fig. 5. Remittance’s own short-term lags and rolling statistics (particularly `Remittance_roll_mean3` and `Remittance_lag12`) rank among the top features for all three models, and exchange rate lags — especially

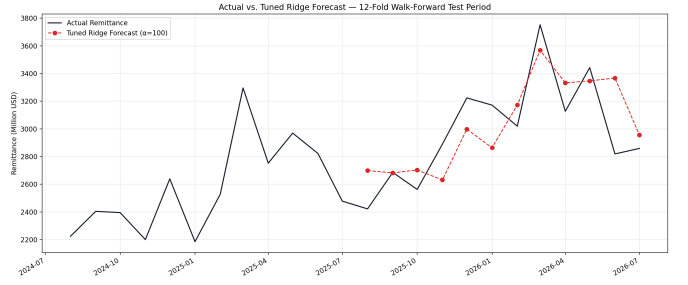


Fig. 4. Actual vs. tuned-Ridge-forecast remittance (million USD) across the 12 walk-forward test months, with the pre-test 12 months of history shown for context.

`ExchangeRate_lag1`, `ExchangeRate_lag2`, and `ExchangeRate_lag12` — consistently appear in the top few features as well, in some cases (Random Forest) ranking above every remittance-lag feature. Inflation lags are present in every model’s top 10 but are generally ranked lower than the leading remittance and exchange-rate features, with LightGBM the partial exception, where `Inflation_lag6` ranks second overall. This is broadly consistent with the lagged cross-correlation pattern observed in the exploratory analysis (Section III-C), where exchange rate showed the strongest short-lag relationship with remittance and inflation the weakest, and is consistent with the Sri Lanka remittance study’s finding that exchange rate dynamics are a dominant short-term driver of remittance fluctuations relative to domestic inflation [9].

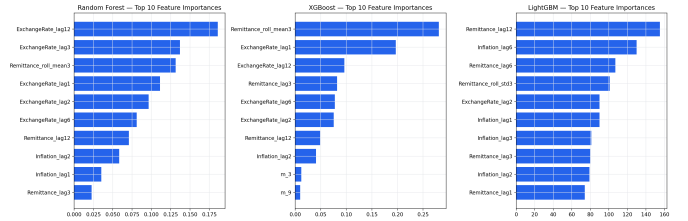


Fig. 5. Three side-by-side horizontal bar charts showing the top-10 feature importances for Random Forest, XGBoost, and LightGBM respectively.

VII. DISCUSSION

A. Why Regularized Linear Models Outperform Trees and Deep Learning Here

The consistent advantage of Ridge Regression over Random Forest, XGBoost, and LightGBM across both evaluation schemes (Tables I–II) is best explained by sample size. After lag-feature construction, roughly 190 monthly observations remain for training, which is a small sample for high-variance ensemble learners that typically require substantially more data to reliably estimate stable split points across many correlated lag features. A regularized linear model, in contrast, has an inductive bias well suited to a feature set dominated by highly collinear lagged versions of the same three underlying series, and the hyperparameter sensitivity analysis (Table III)

shows that even stronger regularization than the library default continues to improve performance monotonically — direct evidence that variance reduction, not additional model flexibility, is what this dataset rewards. This finding echoes the Sri Lanka remittance study, where Ridge Regression was likewise found to be the standout performer among a broader comparison of methods for a comparably sized monthly remittance series [9].

B. The Value of Walk-Forward Validation

The ranking shift between Table I and Table II — most notably XGBoost’s improvement from worst-tier to mid-tier performance — demonstrates concretely why a single fixed train/test split should not be trusted for model selection on a short monthly macroeconomic series. With only 12 held-out points, a single unusually large or small remittance month (for example, an Eid-timing shift between calendar years) can dominate the aggregate error for models that are otherwise reasonably well-specified. The 12-fold expanding-window evaluation used here produces 12 independent one-step-ahead test points per model rather than one aggregate window, giving a substantially more trustworthy basis for model selection, at the cost of additional computation from repeated refitting.

C. Relation to Prior Literature

Our finding that exchange rate lags contribute strongly to remittance prediction (Section VI-E) is consistent with the long-run positive relationship between exchange rate and remittance inflow documented for Bangladesh using gravity-model methodology [5], and with the general finding that exchange rate dynamics tend to dominate short-term remittance fluctuations relative to inflation in comparable South Asian contexts [9]. Unlike the classical ARIMA-based Bangladesh remittance literature [1]–[4], which forecasts remittance from its own history alone, our multivariate feature set allows this relative contribution to be quantified directly rather than only tested for statistical significance in a separate regression.

D. Limitations

Several limitations should be noted. First, the dataset spans roughly 16 years of monthly data, which, after lag-feature construction, leaves a comparatively small effective training sample for the deep learning models; LSTM and GRU may perform more competitively on a longer or higher-frequency series. Second, walk-forward validation was limited to 12 folds and a fixed 12-month lookback window for computational tractability; a longer walk-forward horizon (e.g., 24–36 folds) would further strengthen the reliability of the model ranking. Third, hyperparameter sensitivity analysis for the tabular models (Tables III–IV) was conducted on a single fixed split rather than under walk-forward validation, for the same computational reason; only the single best Ridge configuration was subsequently re-validated under the full walk-forward scheme (Table VI). Finally, the exchange rate and inflation series used here are themselves imperfectly forecastable, and this study assumes their historical values are available up to

the forecast origin — a genuine multi-step-ahead deployment would additionally need to forecast these exogenous inputs or treat them as forecasts-of-forecasts, which was outside the scope of this study.

VIII. CONCLUSION AND FUTURE WORK

In this paper, the authors introduced a multivariate machine learning framework for forecasting monthly remittance inflow to Bangladesh based on four trivial baselines and seven ML/DL models, with a single train/test split and 12-fold walk forward validation. After tuning the hyperparameters (with $\alpha = 100$), Ridge Regression also had the best overall walk-forward performance with RMSE = 246.28 and MAPE = 7.07%. This finding complements the feature importance analysis which indicates that exchange rate lags are strong contributors, along with remittance lags, to remittance prediction, and is similar to recent similar remittance forecasting studies in South Asia. The walk-forward evaluation further confirmed that rankings based on a single fixed split might be misleading for short series of monthly macroeconomic variables, which is further evidence of the need for rolling-origin validation in this context.

Future work will involve: (i) a longer validation window (24 - 36 folds) to further stabilize the model ranking; (ii) an explicit Eid-proximity feature derived from Islamic lunar calendar in addition to fixed-month dummy variables, to improve the ability to capture the drifting seasonal peak that was identified in the exploratory analysis; (iii) an ensemble of tuned Ridge Regression and GRU, due to their different error structures; (iv) an extension of the framework to jointly forecast the exchange rate and inflation itself, e.g., by a VAR/VARX baseline or multi-output neural network, in order to not require exogenous inputs known in advance for genuine multi-step-ahead forecasting; and (v) a longer horizon for the walk-forward validation, which could be pushed further with computational costs.

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